FourMomentum
-energy: float
-x_momentum: float
-y_momentum: float
-z_momentum: float
-rest_mass: float
-FourMomentum()
-FourMomentum(energy_in: float, px_in: float, py_in: float, pz_in: float, mass_in: float)
-validate_infinite(x: float): void <<static>>
-validate_mass(mass_in: float): void <<static>>
-compute_energy(mass_in: float, px_in: float, py_in: float, pz_in: float): float <<static>>
-compute_mass(energy_in: float, px_in: float, py_in: float, pz_in: float): float <<static>>
+~FourMomentum()
+set_px(px_in: float&): void
+set_py(py_in: float&): void
+set_pz(pz_in: float&): void
+get_E(): float&
+get_px(): float&
+get_py(): float&
+get_pz(): float&
+four_momentum_from_mass(mass_in: float, px_in: float, py_in: float, pz_in: float): FourMomentum <<static>>
+four_momentum_from_energy(energy_in: float, px_in: float, py_in: float, pz_in: float): FourMomentum <<static>>
+default_four_momentum(): FourMomentum <<static>>
+operator<<(os: ostream&, fm: FourMomentum&)
Lepton
#mass: float
#relative_charge: int
#momentum_ptr: unique_ptr<FourMomentum>
#is_antiparticle: bool
#base_info: void
+Lepton() = delete
+Lepton(mass_in: float, charge_in: int, px_in: float, py_in: float, pz_in: float, is_anti: bool = false)
+~Lepton() <<virtual>>
+Lepton(const Lepton&)
+Lepton(Lepton&&)
+operator=(const Lepton&): Lepton&
+operator=(Lepton&&): Lepton&
+get_four_momentum(): FourMomentum&
+get_mass(): float&
+get_charge(): float&
+info(): void <<virtual>>
+has_decayed(): bool <<virtual>>
+sum_momenta(lep1: Lepton&, lep2: Lepton&): FourMomentum <<friend>>
+dot_product_momenta(lep1: Lepton&, lep2: Lepton&): float <<friend>>
Electron
-ELECTRON_MASS: float = 0.511 <<static>>
-ELECTRON_CHARGE: int = -1 <<static>>
-detector_length: float
-validate_detector_length(detector_length_in: float): void <<static>>
+Electron()
+Electron(px_in: float, py_in: float, pz_in: float, detector_length_in: float, is_anti: bool = false)
+set_detector_length(detector_length_in: float&): void
+get_detector_length(): float&
+info(): void <<override>>
+has_decayed(): bool <<override>>
+shower_depth_in_tungsten(): float
+shower_contained(detector_length_in: float): string
Muon
-MUON_MASS: float = 105.7 <<static>>
-MUON_CHARGE: int = -1 <<static>>
-MUON_MEAN_LIFETIME: float = 2.2e-6 <<static>>
-MAX_TIME: float = 1.0 <<static>>
-time_in_dector: double = 0
-validate_time(t: float): float <<static>>
+Muon()
+Muon(px_in: float, py_in: float, pz_in: float, time_in: float, is_anti: bool=false)
+set_time_in_detector(time_in: float&): void
+get_time_in_detector(): float&
+info(): void <<override>>
+has_decayed(): bool <<override>>
Tau
-TAU_MASS: float = 1777 <<static>>
-TAU_CHARGE: int = -1 <<static>>
-TAU_MEAN_LIFETIME: float = 2.9e-13 <<static>>
-MAX_TIME: float = 1.0 <<static>>
-PROB_HADRONIC: float = 0.648 <<static>>
-PROB_ELECTRON: float = 0.178 <<static>>
-PROB_MUON: float = 0.174 <<static>>
-time_in_dector: double = 0
-decay_mode: DecayMode
-validate_time(t: float): float <<static>>
-generate_decay_mode(): DecayMode <<static>>
+Tau()
+Tau(px_in: float, py_in: float, pz_in: float, time_in: float, is_anti: bool=false)
+set_time_in_detector(): void
+get_time_in_detector(): float&
+get_decay_mode(): string
+info(): void <<override>>
+has_decayed(): bool <<override>>
<<enumeration>>
DecayMode
Hadron
Electron
Muon
Neutrino
-NEUTRINO_MASS: float = 0 <<static>>
-NEUTRINO_CHARGE: int = -1 <<static>>
-flavour(): NeutrinoFlavour
+Neutrino(flavour_in: NeutrinoFlavour)
+Neutrino(px_in: float, py_in: float, pz_in: float, flavour_in: NeutrinoFlavour, is_anti: bool=false)
+set_flavour(nf: NeutrinoFlavour): void
+get_flavour(): string
get_generation(): int
+info(): void <<override>>
+has_decayed(): bool <<override>>
<<enumeration>>
NeutrinoFlavour
Electron
Muon
Tau